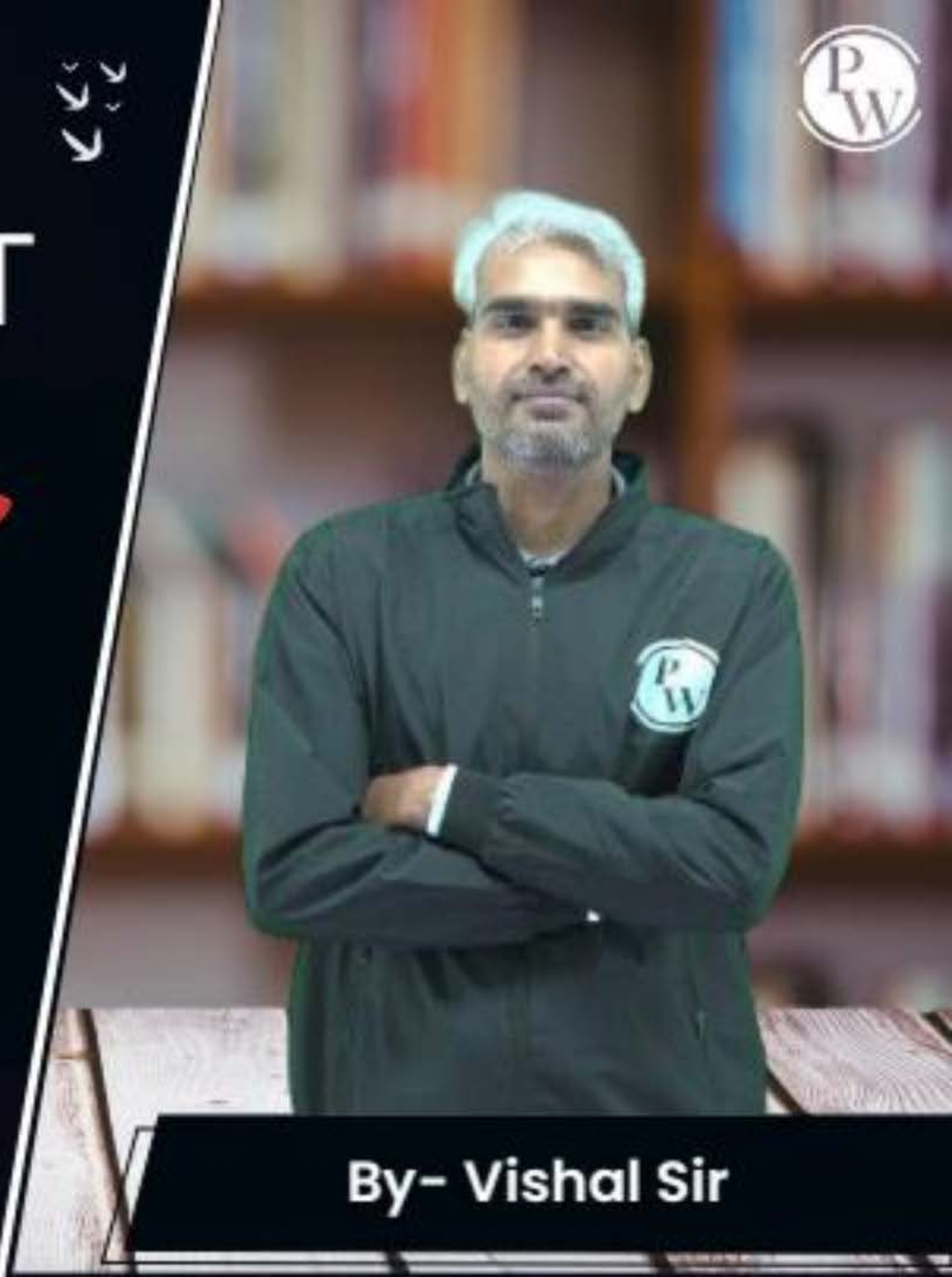
Computer Science & IT



Discrete Mathematics

Combinatorics

Lecture No. 02



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Recap of Previous Lecture



- **Topic** Introduction to recurrence relation
- **Topic** Formation of recurrence relation
- **Topic** Solution of recurrence relation using substitution method

Slide

Topics to be Covered



Topic Method of characteristic roots

Topic Method of undetermined coefficient

Slide



Topic : Linear recurrence relation

The recurrence relation of the form

$$\lambda_0 a_n + \lambda_1 a_{n-1} + \lambda_2 a_{n-2} + \cdots + \lambda_k a_{n-k} = f(n) \quad \text{— eqn (1)}$$

{where $\lambda_0, \lambda_1, \cdots \lambda_k$ are Constants}

is called a linear recurrence relation

In equation (1) if $f(n) = 0$, then recurrence relation is called homogeneous linear recurrence relation

In equation (1) if $f(n) \neq 0$, then it is call non-homogeneous linear recurrence relation



Topic : Method of characteristic roots

Consider the following linear recurrence relation

$$\lambda_0 a_n + \lambda_1 a_{n-1} + \lambda_2 a_{n-2} + \cdots + \lambda_k a_{n-k} = f(n) \quad \text{—eqn ①}$$

{ where $\lambda_0, \lambda_1, \ldots$ are Constants }

Put Put $n := n+k$ in eqn ①

if we put $n := n+k$ in function $f(n)$ then we will get a new function let's call it

i.e Equation becomes,

$$\lambda_0 a_{n+k} + \lambda_1 a_{n+k-1} + \lambda_2 a_{n+k-2} + \cdots + \lambda_k a_n = F(n) \quad \text{—eqn ②}$$

Consider the following shift operator

$$a_{n+i} = E^i \cdot a_n$$

i.e.

$$a_n = E^0 \cdot a_n = a_n$$

$$a_{n+0}$$

$$a_{n+1} = E^1 \cdot a_n$$

$$a_{n+2} = E^2 \cdot a_n$$

$$\vdots$$

$$a_{n+k} = E^k \cdot a_n$$

Our eqn ② is

$$\lambda_0 a_{n+k} + \lambda_1 a_{n+(k-1)} + \lambda_2 a_{n+(k-2)} + \cdots + \lambda_k a_n = F(n) \quad \text{— eqn ②}$$

Put the values from shift operator in to eqn ②

$$\lambda_0 \cdot E^k a_n + \lambda_1 E^{k-1} a_n + \lambda_2 \cdot E^{k-2} a_n + \cdots + \lambda_k a_n = F(n)$$

$$a_n \left[\lambda_0 \cdot E^k + \lambda_1 \cdot E^{k-1} + \lambda_2 \cdot E^{k-2} + \cdots + \lambda_k\right] = F(n)$$

it is a polynomial of degree = k

lets assume $\phi(E) = \left[\lambda_0 E^k + \lambda_1 \cdot E^{k-1} + \cdots + \lambda_k\right]$

$$\equiv \boxed{a_n \cdot \phi(E) = F(n)} \quad \text{— eqn ③}$$

Characteristic Equation

In Equation (3) $\phi(E)$ is Polynomial of degree 'k'

We say that $\phi(t) = 0$ is the Characteristic Equation

Let the roots of Characteristic Equation are

$t_1, t_2, t_3, \ldots\ldots, t_k$ { these are called Characteristic roots }

Polynomial of deg 'k'
∴ 'k' roots

Complementary Function (C.F) :→

$(a_n^{(H)})$ Complementary function is the solution w.r.t. homogeneous part of the linear recurrence relation
{i.e. w.r.t. $f(n) = 0$}

Particular Solution (P.S.)

$(a_n^{(P)})$ Particular solution is the solution w.r.t. non-homogeneous part of linear recurrence relation
{i.e. when $f(n) \neq 0$}

Complete Solution :-

Complete Solution of linear recurrence relation = (Complementary function + Particular Solution)

$$a_n = a_n^{(H)} + a_n^{(P)}$$

$$a_n - 3a_{n-1} + 2a_{n-2} = 2^n$$

Let $a_n^{(H)}$ is C.F. { i.e Soluⁿ w.r.t. homogeneous Part)
i.e. when $f(n) = 0$

Let $a_n^{(P)}$ is P.S. { i.e Soluⁿ w.r.t. non-homogeneous Part
i.e. when $f(n) \neq 0$

$$a_n^{(H)} - 3a_{n-1}^{(H)} + 2a_{n-2}^{(H)} = 0$$

$$a_n^{(P)} - 3a_{n-1}^{(P)} + 2a_{n-2}^{(P)} = 2^n$$

$$\left(a_n^{(H)} + a_n^{(P)}\right) - 3\left(a_{n-1}^{(H)} + a_{n-1}^{(P)}\right) + 2\left(a_{n-2}^{(H)} + a_{n-2}^{(P)}\right) = 2^n$$

a_n a_{n-1} a_{n-2}

Complete Solution (a_n) :-

$$a_n = CF + P.S.$$

(or)

$$a_n = a_n^{(H)} + a_n^{(P)}$$

$$a_n \cdot \phi(E) = F(n) \quad \text{—— eq}^n \ (3)$$

$$a_n = \frac{F(n)}{\phi(E)}$$

Particular solution = Soln of $\left(\frac{F(n)}{\phi(E)}\right)$

if $f(n) = 0$ in eq^n (1), then $F(n) = 0$ in eq^n (3)

∴ if linear recurrence relation is homogeneous linear recurrence rel^n, then $F(n) = 0$

∴ Particular Soln = Soln of $\left(\frac{F(n)}{\phi(E)}\right) = 0$

Note:- If given recurrence relation is a homogeneous linear recurrence relation {i.e, if $f(n)=0$}, then Complementary Function alone will be the solution of that recurrence relation

Because,
When $f(n)=0$,
then Particular Solution = 0

Rules to write Complementary Function

Characteristic Roots	Complementary function
* (1) When all roots are real and distinct. i.e; $t_1, t_2, t_3, \ldots t_k$	(1) $C_1(t_1)^n + C_2(t_2)^n + C_3(t_3)^n + \cdots + C_k(t_k)^n$ Where $C_1, C_2, \ldots C_k$ are Unknown Constants.
(2) When all roots are real and two roots are equal i.e; $t_1, t_1, t_3, t_4, \ldots t_k$	(2) $(C_1 + C_2 n)(t_1)^n + C_3(t_3)^n + \cdots + C_k(t_k)^n$ Where $C_1, C_2, \ldots C_k$ are Unknown Constants.
(3) When all roots are real and three roots are equal i.e; $t_1, t_1, t_1, t_4, \ldots t_k$	(3) $(C_1 + C_2 n + C_3 n^2)(t_1)^n + C_4(t_4)^n + \cdots + C_k(t_k)^n$ Where $C_1, C_2, \ldots C_k$ are Unknown Constants.

* Note:- We can write the Complementary Function Even when some roots are Complex

{ Not required w.r.t. GATE }

Particular Solution

We need to find the particular solution only when $f(n) \neq 0$

Particular Solution

From Eq^{n} (3)

it is the function obtained after replacing 'n' by 'n+k' in F(n)

$$a_n \cdot \phi(E) = F(n)$$

$$a_n = \frac{F(n)}{\phi(E)}$$

Particular Solution { i.e $a_n^{(P)}$ } is the Solution of $\frac{F(n)}{\phi(E)}$

$$P.S. = a_n^{(P)} = \text{Solution of } \left(\frac{F(n)}{\phi(E)}\right)$$

* Rules for writing the particular Solution when $F(n) = (b)^n$

Exponential function with base 'b'

Particular Solution (P.S) = $a_n^{(P)}$ = Solution of $\left(\frac{(b)^n}{\phi(E)}\right)$

Case (1) When 'b' is not a Characteristic root {i.e. $\phi(b) \neq 0$}
in that case

$$P.S = \frac{(b)^n}{\phi(b)}$$

* Rules for writing the particular Solution when $F(n) = \underbrace{(b)^n}_{\text{Exponential function with base 'b'}}$

Particular Solution (P.S) = $a_n^{(P)}$ = Solution of $\left(\frac{(b)^n}{\phi(E)}\right)$

Case (2) When 'b' is a Characteristic root with multiplicity = m {i.e 'b' is a root but 'm' times}

i.e $\phi(b) = 0$ & $\phi(E) = (E-b)^m \cdot \psi(E)$ where $\psi(b) \neq 0$

in that case

$$P.S = a_n^{(P)} = \text{Soln of } \left(\frac{(b)^n}{\phi(E)}\right) = \text{Soln of } \left(\frac{(b)^n}{(E-b)^m \psi(E)}\right) = \boxed{\frac{1}{\psi(b)} \cdot {}^nC_m \cdot (b)^{n-m}}$$

Q:- Find the solution of the recurrence relation, $a_n = 3 \cdot a_{n-1}$ where $\boxed{a_0 = 1}$

$\Downarrow$

$a_n - 3a_{n-1} = 0$ ← homogeneous

Put $n-1 = n$

$a_{n+1} - 3a_n = 0$

↳ Use shift operator

i.e $a_{n+i} = E^i \cdot a_n$

$E^1 a_n - 3 \cdot E^0 \cdot a_n = 0$

$E \cdot a_n - 3E \cdot a_n = 0$

$a_n (E-3) = 0$

$\phi(E)$, $F(n)$

$\phi(E) = E - 3$

and Ch. eqn is $\phi(t) = 0$

∴ Ch Eqn is $\boxed{t-3=0}$

↳ Hence Ch. root is '3'

∴ Complementary function

$\boxed{CF = C_1 \cdot (3)^n}$

$F(n) = 0$

∴ Particular Soln = 0

Complete Soln

$a_n = CF + PS \to 0$

$a_n = C_1 (3)^n$

We know $a_0 = 1$

$a_0 = 1 = C_1 \cdot (3)^0$

$\Rightarrow \boxed{C_1 = 1}$

Complete Soln $a_n = 1 \cdot (3)^n \Rightarrow \boxed{a_n = 3^n}$

Q:- Find the solution of the recurrence relation, $a_n - 7a_{n-1} + 12a_{n-2} = 0$ where $a_0 = 1$, $a_1 = 2$

Put $n = n+2$ & use shift operator

Then $E^2 a_n - 7Ea_n + 12a_n = 0$

$a_n(\underbrace{E^2 - 7E + 12}_{\phi(E)}) = 0$ ← $F(n)$

Ch eqn $\phi(t) = 0$

$t^2 - 7t + 12 = 0$,

Ch roots, $\Rightarrow 3, 4$

$$CF = C_1(3)^n + C_2(4)^n$$

$PS = 0$

$a_n = CF + \cancel{PS} \to 0$

$a_n = C_1(3)^n + C_2(4)^n$

$a_0 = 1$ & $a_1 = 2$

$\therefore a_0 = 1 : C_1(3)^0 + C_2(4)^0$

$1 : C_1 + C_2$

$a_1 = 2 : C_1(3)^1 + C_2(4)^1$

$2 : 3C_1 + 4C_2$

$C_1 = 2$

$C_2 = -1$

Hence

$$a_n = 2(3)^n - 1(4)^n$$

Q:- Find the solution of the recurrence relation, $a_n = 2a_{n-1} - 1$ where $a_1 = 2$

$$a_n - 2a_{n-1} = -(1)^n$$

Put $n = n+1$ & use shift operator

$$E \cdot a_n - 2a_n = -(1)^{n+1}$$

$$a_n(E-2) = -(1)^n$$

$\phi(E)$, $F(n)$

Ch. eqn.: $t - 2 = 0$

Ch. root: $t_1 = 2$ ✓

$$C.F. = C_1(2)^n$$

$$a_n(E-2) = -(1)^n \quad F(n)$$

$$P.S. = \text{Soln of } \left(\frac{-(1)^n}{(E-2)}\right) = -\left\{\text{Soln of } \frac{(1)^n}{(E-2)}\right\}$$

base $b = 1$

'1' is not a root

$\phi(E)$

$$\therefore P.S. = -\left\{\frac{(1)^n}{\phi(1)}\right\} = -\left\{\frac{1}{1-2}\right\}$$

$\phi(b)$

$$P.S. = +1$$

Complete Soln $a_n = C.F. + P.S.$

$$a_n = C_1 \cdot (2)^n + 1$$

$a_1 = 2 = C_1 \cdot (2)^1 + 1 \Rightarrow C_1 = \frac{1}{2}$

$$\therefore a_n = 2^{n-1} + 1$$

Q:- Find the solution of the recurrence relation.

$a_n = 2a_{n-1} + 2^n$ $\boxed{a_0 = 1}$

$a_n - 2a_{n-1} = 2^n$

$E.a_n - 2a_n = (2)^{n+1}$

$a_n(E-2) = 2.(2)^n$

$\Downarrow$

$a_n = \dfrac{2.(2)^n}{(E-2)}$

$\phi(E)$

CF = $C_1.(2)^n$ ($t = 2$)

'2' is a root with multiplicity '1'

PS = Soln of $\left(\dfrac{2.(2)^n}{\phi(E)}\right)$ base $b = 2$

$= 2.\left\{\text{Soln of } \left(\dfrac{(2)^n}{(E-2)^1 \cdot 1}\right)\right\}$ $\psi(E)$

$= 2.\left\{\dfrac{1}{\psi(2)} \cdot {}^{n}C_1 (2)^{n-1}\right\}$ (m = multiplicity = 1; b; $m=1$; $b=2$)

$= 2.\{n.2^{n-1}\} \Rightarrow \boxed{PS = n.2^n}$

$\psi(E) = 1$ it is a Constant fn. $\therefore \psi(2) = 1$

$a_n = CF + PS$

$a_n = C_1(2)^n + n.2^n$

$a_0 = 1 = C_1.(2)^0 + 0.2^0$

$\boxed{1 = C_1}$

$\therefore a_n = 2^n + n.2^n$

$= 2^n.(n+1)$

Q:- Find the soln of recurrence relation,

$$a_n - 2a_{n-1} + a_{n-2} = 0$$

$a_0 = 1$
$a_1 = 2$

$\phi(E) = E^2 - 2E + 1$

Ch. roots :- 1, 1

$CF = (C_1 + C_2 n)(1)^n$

$PS = 0$

$a_n = CF + PS$

$a_n = (C_1 + C_2 n)(1)^n$

$\boxed{a_n = C_1 + C_2 n}$ => $a_0 = 1$ & $a_1 = 2$

$\therefore C_1 = ? = 1 \quad C_2 = ? = 1$

$\therefore \boxed{a_n = (1 + n) = (n + 1)}$

Q:- Find the solution of recurrence relation

$$a_n - 3a_{n-1} + 2a_{n-2} = 2^n$$

$a_0 = 1$
$a_1 = 2$

$\phi(E) = E^2 - 3E + 2$

$F(n) = 2^{n+2}$

Ch. roots $= 2, 1$

$$CF = C_1(2)^n + C_2$$

PS = Soln of $\left(\frac{F(n)}{\phi(E)}\right)$

$= \text{Soln of } \left(\frac{4 \cdot 2^n}{(E-2)(E-1)}\right)$

$= 4 \cdot \text{Soln of } \left(\frac{2^n}{(E-2)^1(E-1)}\right)$

$b = 2$

$\psi(E)$

'2' is a root with multiplicity $= 1$

$$PS = 4 \times \left\{ \frac{1}{\psi(2)} \cdot n C_1 (2)^{n-1} \right\}$$

$$= 4 \times \left\{ \frac{1}{(2-1)} \cdot n \cdot 2^{n-1} \right\}$$

$$PS = n \cdot 2^{n+1}$$

Complete Soln

$a_n = CF + PS$

$= C_1(2)^n + C_2 + n \cdot 2^{n+1}$

$C_1 = -3$, $C_2 = 4$

$$a_n = -3(2)^n + 4 + n \cdot 2^{n+1}$$

Q:. Find the Solution of recurrence relation

$$T(3^k) = 2 \cdot T(3^{k-1}) + 1$$

let $T(1) = 1$

let $T(3^k) = a_k$

$\therefore$ Eqn becomes

$$a_k = 2 \cdot a_{k-1} + 1$$

$$a_k - 2\,a_{k-1} = (1)^k$$

$\phi(E) = (E-2)$

$F(n) = (1)^{k+1} = (1)^k$

Ch. roots $\Rightarrow$ "2"

$$Cf = C_1 \cdot (2)^k$$

PS = Soln of $\frac{(1)^k}{(E-2)}$ — base $k = 1$, not a root

$\therefore PS = \frac{(1)^k}{(1-2)}$ — $\phi(1)$

$$PS = -1$$

Complete Soln

$a_k = CF + PS$

$a_k = C_1 \cdot (2)^k - 1$

$a_k = T(3^k) = C_1 (2)^k - 1$

$T(1) = 1$

$3^k = 1$ at $k = 0$

$\therefore T(1) = 1 = C_1 (2)^0 - 1$

$$C_1 = 2$$

$$T(3^k) = 2 \cdot (2)^k - 1 = 2^{k+1} - 1$$

* **Q:-** Find the solution of recurrence relation

$$a_n^2 - 2 \cdot a_{n-1}^2 = 1 \qquad a_1 = 2$$

it is not a linear recurrence relation, but we can convert it into linear recurrence relation by replacing $a_n^2 = x_n \Rightarrow a_n = \pm\sqrt{x_n}$

∴ Eqn becomes

$$x_n - 2 \cdot x_{n-1} = 1$$

$x_n - 2x_n = 1$

$\phi(E) = E - 2$

$F(n) = (1)^{n+1} = (1)^n$

Ch root = 1 '2'

$CF = C_1 \cdot (2)^n$

PS: Soln of $\frac{(1)^n}{(E-2)}$ — 'b' = 1, 'b' is not a root

$\therefore PS = \frac{(1)^n}{(1-2)} = -1$

Complete Soln:

$x_h = CF + PS$

$$\boxed{x_h = C_1(2)^n - 1}$$

i.e. $a_n^2 = C_1(2)^n - 1$

$a_1 = 2 \Rightarrow (a_1)^2 = C_1(2)^1 - 1$

$= (2)^2 = C_1(2)^1 - 1 \Rightarrow \boxed{C_1 = 5/2}$

$a_n^2 = \frac{5}{2} \cdot (2)^n - 1$

$a_n^2 = 5 \cdot (2)^{n-1} - 1$

$$a_n = \pm\sqrt{5(2)^{n-1} - 1}$$

Q:- Find the solution of recurrence relation

$$\sqrt{a_n} - \sqrt{a_{n-1}} - 2\sqrt{a_{n-2}} = 0$$

$a_0 = 1$
$a_1 = 1$

Replace $\sqrt{a_n}$ by x_n

∴ Eqⁿ becomes

$$x_n - x_{n-1} - 2x_{n-2} = 0$$

$$x_n - x_{n-1} - 2x_{n-2} = 0$$

$\phi(E) = E^2 - E + 2$

$F(n) = 0$

Ch. roots $= 2, -1$

$\therefore CF = C_1 \cdot (2)^n + C_2 (-1)^n$

$PS = 0$

Complete soln.

$x_n = CF + PS$

$x_n = \sqrt{a_n} = C_1 (2)^n + C_2 (-1)^n$

$a_0 = 1 \Rightarrow \sqrt{a_0} = C_1 \cdot (2)^0 + C_2 (-1)^0 = 1$

$a_1 = 1 \Rightarrow \sqrt{a_1} = C_1 (2)^1 + C_2 (-1)^1 = 1$

$C_1 = \frac{2}{3}$ & $C_2 = 1/3$

$\therefore \sqrt{a_n} = \frac{2}{3}(2)^n + \frac{1}{3}(-1)^n$

$$a_n = \left(\frac{2}{3}(2)^n + \frac{1}{3}(-1)^n\right)^2$$

Divide & Conquer Relation

Divide & Conquer Relation

A relation of the type,

$$T(n) = c \cdot T\left(\frac{n}{d}\right) + f(n)$$

is called divide & conquer relation

where c & d are constants

* We can convert the given recurrence relation in to linear recurrence relation

Put $n = d^k$

∴ Eqn becomes

$$T(d^k) = c \cdot T\left(\frac{d^k}{d}\right) + g(k)$$

Function after replacing 'n' by d^k in $f(n)$

$$T(d^k) = c \cdot T(d^{k-1}) + g(k)$$

Put $T(d^k) = x_k$

$$\therefore \boxed{x_k = c \cdot x_{k-1} + g(k)}$$

it is linear recurrence relation

Q:- Find the solution of the recurrence relation

$$T(n) = 7 \cdot T\left(\frac{n}{3}\right) + 2n$$

where $T(1) = 2$

Put $n = 3^k$

$$\therefore \quad T(3^k) = 7 \cdot T\left(\frac{3^k}{3}\right) + 2 \cdot (3)^k$$

$$T(3^k) = 7 \cdot T(3^{k-1}) + 2 \cdot (3)^k$$

Put $T(3^k) = x_k$

$$\therefore \quad x_k = 7 \cdot x_{k-1} + 2(3)^k$$

$$\boxed{x_k - 7 \cdot x_{k-1} = 2 \cdot (3)^k}$$

$\phi(E) = (E-7)$

$F(n) : 2 \cdot (3)^{k+1} = 6 \cdot (3)^k$

Ch. root $\Rightarrow$ '7'

$$\boxed{CF = C_1 \cdot (7)^k}$$

PS = Soln of $\left(\frac{F(n)}{\phi(E)}\right)$

$= \text{Soln of} \left(\frac{6 \cdot (3)^k}{(E-7)}\right)$

$= 6 * \text{Soln of} \left(\frac{(3)^k}{(E-7)}\right)$

base $b = 3$

'3' is not a root

$$PS = 6 * \left(\frac{(3)^k}{(3-7)}\right)$$

$$\boxed{PS = \frac{-3}{2} \cdot (3)^k}$$

$\therefore$ Complete Soluh

$$x_k = C_1(7)^k - \frac{3}{2}\cdot(3)^k$$

$$x_k = T(3^k) = C_1\cdot(7)^k - \frac{3}{2}\cdot(3)^k$$

$T(3^k) = T(1)$ at $k = 0$

$$T(1) = T(3^0) = C_1\cdot(7)^0 - \frac{3}{2}(3)^0$$

$$1(1) = 2 = C_1 - \frac{3}{2}$$

$$C_1 = \frac{7}{2}$$

$$T(3^k) = \frac{7}{2}\cdot(7)^k - \frac{3}{2}\cdot(3)^k$$

we know $3^k = n$

$$\therefore \log_3(3)^k = \log_3 n$$

$$k\cdot\log_3(3) = \log_3 n$$

($\log_3(3) = 1$)

$$k = \log_3 n$$

$$\therefore T(n) = \frac{7}{2}(7)^{\log_3 n} - \frac{3}{2}(n)$$

(3^k → n; $3k$ → n)

$$T(n) = \frac{7}{2}\cdot(7)^{\log_3 n} - \frac{3}{2}n$$

Note:-

$$\text{Particular Solution of } \left(\frac{(a)^n + (b)^n}{\phi(E)}\right) = \text{Particular Solution of } \left(\frac{(a)^n}{\phi(E)} + \frac{(b)^n}{\phi(E)}\right)$$

$$= \text{PS of } \left(\frac{(a)^n}{\phi(E)}\right) + \text{PS of } \left(\frac{(b)^n}{\phi(E)}\right)$$

Q: Find the Soln of recurrence relⁿ

$$T(n) = 2 \cdot T\left(\frac{n}{2}\right) + (n+1)$$

$d = 2$

Where $T(1) = 0$

Put $n = 2^k$

$\therefore T(2^k) = 2 \cdot T(2^k) + 2^k + 1$

Put $T(2^k) = x_k$

$\therefore x_k = 2 \cdot x_{k-1} + 2^k + 1$

$$x_k - 2 \cdot x_{k-1} = (2)^k + (1)^k$$

$\phi(E) = (E-2)$

$F(n) = (2)^{k+1} + (1)^{k+1}$

Characteristic root $= 2$

$\therefore CF = C_1(2)^k$

P.S = Soln of $\left(\frac{F(n)}{\phi(E)}\right)$

$= \text{Soln of } \left(\frac{(2)^{k+1} + (1)^{k+1}}{(E-2)}\right)$

$= \text{Soln of } \left(\frac{2 \cdot (2)^k}{(E-2)}\right) + \text{Soln of } \left(\frac{(1)^k}{(E-2)}\right)$

Q: Find the soln of recurrence relⁿ

$T(n) = 2 \cdot T\left(\frac{n}{2}\right) + (n+1)$ where $T(1) = 0$

PS : Soln of $\left(\frac{2 \cdot (2)^k}{(E-2)}\right)$ + Soln of $\left(\frac{(1)^k}{(E-2)}\right)$

: $2 \times$ Soln of $\left(\frac{(2)^k}{(E-2)^1 \cdot 1}\right)$ + Soln of $\left(\frac{(1)^k}{(E-2)}\right)$

base 'b = 2' for $(2)^k$; base 'b = 1' for $(1)^k$

'2' is a root with multiplicity = 1

$\Psi(E)$ constant function $\therefore \Psi(2) = 1$

'1' is not a root

$$PS = 2 \times \left\{ \frac{1}{\Psi(2)} \cdot {}^{k}C_1 \cdot (2)^{k-1} \right\} + \left\{ \frac{(1)^k}{(1-2)} \right\}$$

($\Psi(2) = 1$)

$$PS: \boxed{k \cdot 2^k - 1}$$

∴ Complete Soln

$$x_k = CF + PS$$

$$x_k = c_1 \cdot (2)^k + k \cdot 2^k - 1$$

$$x_k = \boxed{T(2^k) = c_1 \cdot (2)^k + k \cdot 2^k - 1}$$

We know $2^k = n$

$\therefore k = \log_2 n$

Hence $\boxed{T(n) = c_1 \cdot n + (\log_2 n) \cdot n - 1}$

given $T(1) = 0$

$T(1) = 0 = c_1 \cdot 1 + (\log_2 1) \cdot 1 - 1$ → $\log_2 1 = 0$

$\boxed{c_1 = 1}$

∴ $\boxed{T(n) = n + n(\log_2 n) - 1}$



Topic : Method of undetermined coefficient

- Method of undetermined coefficient can be used to identify the solution of non-homogeneous part of linear recurrence relation when function f(n) is neither a constant nor an exponential function

i.e. (Particular Soln)

i.e $f(n) = C_1 \cdot (1)^n$

Constant

$f(n) = C_2 \cdot (b)^n$

Constant

base

Method of undetermined Coefficient

Consider the following linear recurrence relation

$$\lambda_0 \cdot a_n + \lambda_1 \cdot a_{n-1} + \lambda_2 \cdot a_{n-2} + \cdots + \lambda_k \cdot a_{n-k} = f(n) \quad \text{—eq}^n (1)$$

after replacing 'n-k' by 'n', and after using shift operator Eq^n becomes.

$$\lambda_0 \cdot E^k \cdot a_n + \lambda_1 \cdot E^{k-1} \cdot a_n + \lambda_2 \cdot E^{k-2} \cdot a_n + \cdots + \lambda_k \cdot a_n = F(n)$$

(after replacing 'n' by 'n+k')

i.e $$a_n \left(\lambda_0 \cdot E^k + \lambda_1 \cdot E^{k-1} + \lambda_2 \cdot E^{k-2} + \cdots + \lambda_k \right) = F(n)$$

where $$\phi(E) = \lambda_0 \cdot E^k + \lambda_1 \cdot E^{k-1} + \lambda_2 \cdot E^{k-2} + \cdots + \lambda_k$$

$\therefore$ Eq^n becomes $a_n \cdot \phi(E) = F(n)$

{ where $\phi(t) = 0$ is the ch. eq^n and roots of that eq^n are called ch. roots. Based on ch. roots we will be able to define the Complementary function

Method of undetermined Coefficient

Rules for writing Particular Solution

Function $f(n)$ from Eq^n-①	$\phi(E)$	Particular Solution
① When function $f(n)$ is of the form, $(b)^n \{C_0 n^s + C_1 n^{s-1} + \dots + C_s\}$ (Exponential function; Polynomial of deg $= s$; where $C_0, C_1, \dots C_s$ are Constants)	Case ① when $\phi(b) = 0$, and 'b' is a Characteristic root with multiplicity $= m$	$(b)^n (A_0 n^s + A_1 n^{s-1} + \dots + A_s)(n)^m$ where $A_0, A_1, \dots A_s$ are unknown Coefficient And they will be determined by satisfying the equation
	Case ② When $\phi(b) \neq 0$ i.e, 'b' is not a Characteristic root. {i.e. 'b' is Ch. root with multiplicity $= 0$}	$(b)^n (A_0 n^s + A_1 n^{s-1} + \dots + A_s)(n)^0$ $\equiv (b)^n (A_0 n^s + A_1 n^{s-1} + \dots + A_s)$ where $A_0, A_1, \dots A_s$ are unknown Coefficient

Method of undetermined Coefficient

Rules for writing Particular Solution

Function $f(n)$ from Eq^n ①	$\phi(E)$	Particular Solution
② When function $f(n)$ is of the form, $\{C_0 n^s + C_1 n^{s-1} + \dots + C_s\}$ (Polynomial of degree $= s$) $\equiv (1)^n \{C_0 n^s + C_1 n^{s-1} + \dots + C_s\}$	Case ① when $\phi(1) = 0$, and '1' is a Characteristic root with multiplicity $= m$	$(1)^n (A_0 n^s + A_1 n^{s-1} + \dots + A_s)(n)^m$ where $A_0, A_1, \dots A_s$ are unknown coefficient and they will be determined by satisfying the equation
	Case ② when $\phi(1) \neq 0$ i.e. '1' is not a Characteristic root. {i.e. '1' is Ch. root with multiplicity $= 0$}	$(1)^n (A_0 n^s + A_1 n^{s-1} + \dots + A_s)(n)^0$ $\equiv (1)^n (A_0 n^s + A_1 n^{s-1} + \dots + A_s)$ where $A_0, A_1, \dots A_s$ are unknown coefficient

$C_0, C_1, C_2 \dots$ etc are known Constants

#Q. Find the solution of recurrence relation

$a_n - 3a_{n-1} = n+3$

$f(n) = (1.n+3)$ Polynomial of deg : 1

$f(n) = (1.n+3)$

$= (1)^n . (1.n+3)$

$(b=1)$

'1' is not a characteristic root

i.e. root with multiplicity = 0

$\phi(E) = E-3$

Ch. eqn: $t-3=0$

$\therefore$ Ch root $\Rightarrow$ $t=3$

$\therefore$ $\boxed{CF = C_1.(3)^n}$

$\therefore$ Particular Soln $= a_n^{(P)} = (1)^n . (An+B) . n^0$

$\boxed{a_n^{(P)} = (An+B)}$

Where A & B are unknown Coefficients

$$a_n - 3a_{n-1} = n+3$$

$a_n^{(p)}$ is a particular soln. ∴ it will satisfy the linear recurrence relation

i.e $\boxed{a_n^{(p)} - 3a_{n-1}^{(p)} = n+3}$

We have, $a_n^{(P)} = An + B$

We know, Particular solution must satisfy the given linear recurrence rel^h

given recurrence rel^h

$$a_n - 3a_{n-1} = (n+3)$$

$$\therefore \boxed{a_n^{(P)} - 3 \cdot a_{n-1}^{(P)} = n+3}$$

i.e. $(An+B) - 3 \cdot (A(n-1)+B) = n+3$

Put $n=0 \Rightarrow (A \cdot 0 + B) - 3(A(0-1)+B) = 0+3 \Rightarrow 3A - 2B = 3$

Put $n=1 \Rightarrow (A \cdot 1 + B) - 3(A(1-1)+B) = 1+3 \Rightarrow A - 2B = 4$

$\left.\begin{array}{l} A = -1/2 \\ B = -9/4 \end{array}\right\}$

$\therefore$ Particular Solution $= \left(\frac{-1}{2} n + \left(\frac{-9}{4}\right)\right) \Rightarrow \boxed{a_n^{(P)} = \frac{-n}{2} - \frac{9}{4}}$

Complete Solution

$$a_n = CF + P.S.$$

$$\boxed{a_n = c_1 \cdot (3)^n + \left(\frac{-n}{2} - \frac{9}{4}\right)}$$

← using boundary Condⁿ we will be able to identify Constant "c_1"

#Q. Find the solution of recurrence relation

$$a_n - a_{n-1} = n+2$$

$\phi(E) = E - 1$

Ch. eqn $t - 1 = 0$

Ch. root $t = 1$

'1' is a root with multiplicity = 1

$C.F. = C_1 \cdot (1)^n$

$\boxed{CF = C_1}$

$f(n) = (n+2)$

$= (1)^n \cdot (n+2)$

'1' is a root with multiplicity = 1

$\therefore P.S.(a_n^{(p)}) = (1)^n \cdot (An+B)(n)^1$

$$\boxed{a_n^{(p)} = n \cdot (An+B)}$$

$a_n^{(p)} = P.S = n(An+B)$

and it will satisfy the given recurrance relation

$$a_n - a_{n-1} = n+2$$

$$\therefore \underbrace{n\cdot(An+B)} - \underbrace{(n-1)(A(n-1)+B)} = n+2$$

Put $n=0$, $-(0-1)(A(-1)+B) = 2 \Rightarrow -A+B = 2$

Put $n=1$, $(A+B) - 0 = 2 \Rightarrow A+B = 3$

$\left.\begin{matrix} B = 5/2 \\ A = 1/2 \end{matrix}\right\}$

$\therefore P.S = n\cdot\left(\frac{n}{2}+\frac{5}{2}\right)$

Complete Solu^n $\Rightarrow$ $a_n = C.F + P.S.$

$$\boxed{a_n = C_1 + \frac{n^2}{2} + \frac{5n}{2}}$$

#Q. Find the solution of recurrence relation

$$a_n - 2.a_{n-1} + a_{n-2} = 3n+5$$

$\phi(E) = E^2 - 2E + 1$

$\therefore$ Ch. eqn : $t^2 - 2t + 1 = 0$

Ch. roots : $t_1 = 1,\ t_2 = 1$

'1' is a root with multiplicity = 2

$C.F = (C_1 + C_2 \cdot n)(1)^n$

$\boxed{CF = C_1 + C_2 \cdot n}$

$f(n) = 3n + 5$

$= (1)^n \cdot (3n+5)$

base $b = 1$, & '1' is a root with multiplicity = 2

$P.S : a_n^{(P)} = (1)^n \cdot (An+B) \cdot n^{(2)}$

$\boxed{a_n^{(P)} = n^2 \cdot (An+B)}$

$a_n^{(P)} = n^2(An+B)$

$$a_n - 2a_{n-1} + a_{n-2} = 3n+5$$

$$n^2(An+B) - 2\cdot(n-1)^2\cdot(A(n-1)+B) + (n-2)^2\cdot(A(n-2)+B) = 3n+5$$

We get, $A = 1/2$ ✓

$B = 4$ ✓

$$\therefore PS = n^2\cdot\left(\frac{n}{2}+4\right)$$

Complete Soln

$$a_n = CF + PS. \Rightarrow (C_1 + C_2 n) + \left(\frac{n^3}{2} + 4n^2\right)$$

#Q. Find the solution of recurrence relation

$a_n - a_{n-1} = 2^n.n$

$f(n) = 2^n (n)$

$= 2^n (1.n+0)$

Polynomial of deg = 1

base $b=2$, '2' is not a root

$\phi(E) = E - 1$

Ch. eqn $= t - 1 = 0$

Ch root $\Rightarrow t = 1$

$CF = C_1 (1)^n$

$CF = C_1$

$\therefore PS = a_n^{(P)} = (2)^n (An+B)(n)^0$

$a_n^{(P)} = 2^n (An+B)$

$$a_n^{(p)} = 2^n \cdot (An + B)$$

$$a_n - a_{n-1} = 2^n \cdot n$$

$$2^n \cdot (An + B) - 2^{(n-1)} \cdot (A(n-1) + B) = 2^n \cdot n$$

We get $\begin{cases} A = 2 \\ B = -2 \end{cases}$

$$\therefore PS = 2^n (2n - 2)$$

Complete Soln

$$\therefore a_n = CF + PS = \boxed{C_1 + 2^n \cdot (2n - 2)}$$

#Q. Find the solution of recurrence relation

$$a_n - 3.a_{n-1} = 3^n(n+2)$$

$\phi(E) = E - 3$

$Ch \cdot eq^n = t - 3 = 0$

Ch root : $t = 3$

$$CF : c_1 \cdot (3)^n$$

$f(n) = (3)^n \cdot (n+2)$

base $b = 3$, '3' is a root with multiplicity = 1

$$PS : a_n^{(p)} = (3)^n \cdot (An+B) \cdot n^{(1)}$$

$$a_n^{(p)} = n \cdot 3^n \cdot (An+B)$$

$$a_n^{(p)} = n \cdot 3^n \cdot (An+B)$$

$$a_n - 3a_{n-1} = 3^n(n+2)$$

$$\therefore \quad n \cdot 3^n(An+B) - 3\left\{(n-1) \cdot 3^{(n-1)} \cdot (A(n-1)+B)\right\} = 3^n(n+2)$$

$$\begin{cases} A = 1/2 \\ B = 5/2 \end{cases}$$

$$\therefore PS = n \cdot 3^n\left(\frac{n}{2} + \frac{5}{2}\right) = 3^n \cdot \left(\frac{n^2}{2} + \frac{5n}{2}\right)$$

Complete Soluⁿ $\boxed{a_n = c_1 \cdot (3)^n + 3^n\left(\frac{n^2}{2} + \frac{5n}{2}\right)}$



2 mins Summary

Topic Method of characteristic roots

Topic Method of undetermined coefficient

Slide

THANK - YOU